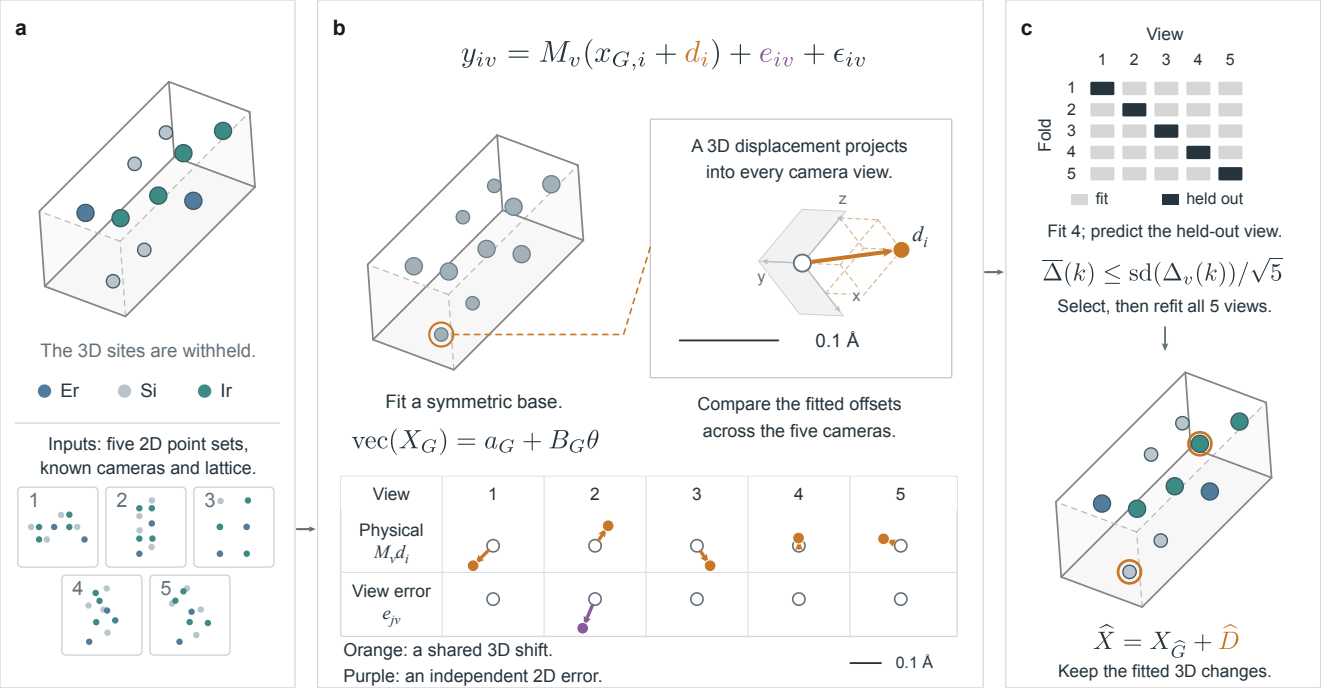

a
The 3D sites are withheld.
Er
Si
Ir
Inputs: five 2D point sets, known cameras and lattice.
1
2
3
4
5
b
$y_{iv} = M_v(x_{G,i} + d_i) + e_{iv} + \epsilon_{iv}$
A 3D displacement projects into every camera view.
z
y
x
d_i
0.1 Å
Fit a symmetric base.
$\mathrm{vec}(X_G) = a_G + B_G\theta$
Compare the fitted offsets across the five cameras.
View
1
2
3
4
5
Physical $M_v d_i$
View error e_{jv}
Orange: a shared 3D shift.
Purple: an independent 2D error.
0.1 Å
c
View
1 2 3 4 5
Fold
1
2
3
4
5
fit
held out
Fit 4; predict the held-out view.
$\overline{\Delta}(k) \le \mathrm{sd}(\Delta_v(k))/\sqrt{5}$
Select, then refit all 5 views.
$\widehat{X} = X_{\widehat{G}} + \widehat{D}$
Keep the fitted 3D changes.